

Questions Bank (Unit-1)

1. Use Warshall's algorithm to calculate transitive Closure of relation R on $A = \{1,2,3\}$ where $R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$
2. Define poset. Draw the Hasse diagram of all divisors of 48. Is it a lattice?
3. Prove using a truth table that $(P \vee Q) \wedge (\neg P \vee R)$ is logically equivalent to $(P \wedge R) \vee (Q \wedge \neg R)$.
4. Apply Warshall's Algorithm to compute the transitive closure of a relation represented by the adjacency matrix below:
$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$
5. Define a poset. Draw the Hasse diagram for the divisors of 60 under the "divides" relation and verify whether that forms a poset or not.
6. Prove by mathematical induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n.
7. Using Warshall's algorithm, find transitive closure of the relation $R = \{(1,4), (2,1), (2,3), (3,1), (3,4), (4,3)\}$ on set $A = \{1, 2, 3, 4\}$.
8. Show that the division relation on set of integers is a partial ordering relation.
9. Draw the Hasse diagram representing the partial ordering $\{(a,b): a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.